

Cryptographic Hashing

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9/1/26

Authentication

- Passwords
- Hopelessness
- Password Managers
- Password attacks
- Password defenses
- Incident response plan!

The beginning of the end of the password

- This might be the last time I have to talk about passwords in the present tense.
 - ■ Matt Kijowski, Sep 12, 2023
 - ■ Matt Kijowski, Jan 16, 2024
 - ■ Matt Kijowski, Sep 5, 2024
 - ■ Matt Kijowski, Jan 28, 2025
 - ■ Matt Kijowski, Sep 2, 2025
 - ■ Matt Kijowski, Jan 22, 2026

What is Authentication

- The act of showing something to be true, genuine, or valid.

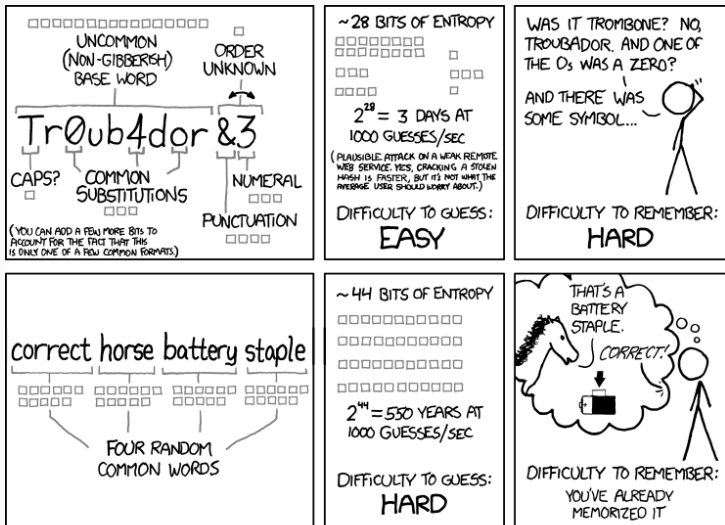
In cybersecurity this usually means

Verifying the identity of a user or process

Passwords

- Most common form of authentication
- “Something you know”
- Different ideas of strong versus weak passwords
- 12345
- Should **NEVER** be stored plain-text
 - salt and hash passwords before storing
 - be sure the transmission / processes for authentication do not leak information
 - probably best not to re-invent the wheel on this one...

CorrectHorseBatteryStaple



THROUGH 20 YEARS OF EFFORT, WE'VE SUCCESSFULLY TRAINED EVERYONE TO USE PASSWORDS THAT ARE HARD FOR HUMANS TO REMEMBER, BUT EASY FOR COMPUTERS TO GUESS.

Password weaknesses

- People
- Weak passwords
- Phishing
- Shoulder surfing
- Leaks (raw or hashed!)
- Dictionaries
- Rainbow Tables
- Brute force
- Side channel attacks!!!
 - Password resets
 - Removal of MFA devices
 - Account recovery
- Bypass attacks
- People

Password Attacks

Generally can be classified into two types:

- Online Password attacks
- Offline Password attacks

Attacks the login interface directly, frequently limited by speed (of network / response from authenticator / input).

- Brute force
- Smarter brute force (dictionary / rainbow tables)
- Shoulder surfing (watching someone enter password)
- Pass the hash (application accepts hashes or passwords)
- Bypass (steal / access an already authenticated system)

Breaking into Matt Kijowski's vehicle

The *ONLINE* attack

I know this is bad, but this attack does NOT require the internet.

9 9 9 9 1 1 1 1 1 3 1 1 1 1 5 1 1 1 1 7 1 1 1 1 9 1 1 1 3
7 1 1 1 3 9 1 1 1 5 3 1 1 1 5 5 1 1 1 5 7 1 1 1 5 9 1 1 1
7 7 1 1 1 7 9 1 1 1 9 3 1 1 1 9 5 1 1 1 9 7 1 1 1 9 9 1 1
3 1 7 1 1 3 1 9 1 1 3 3 3 1 1 3 3 5 1 1 3 3 7 1 1 3 3 9 1
1 3 5 7 1 1 3 5 9 1 1 3 7 3 1 1 3 7 5 1 1 3 7 7 1 1 3 7 9
1 1 3 9 7 1 1 3 9 9 1 1 5 1 3 1 1 5 1 5 1 1 5 1 7 1 1 5 1
5 1 1 5 3 7 1 1 5 3 9 1 1 5 5 3 1 1 5 5 5 1 1 5 5 7 1 1 5
7 5 1 1 5 7 7 1 1 5 7 9 1 1 5 9 3 1 1 5 9 5 1 1 5 9 7 1 1
7 1 5 1 1 7 1 7 1 1 7 1 9 1 1 7 3 3 1 1 7 3 5 1 1 7 3 7 1
1 7 5 5 1 1 7 5 7 1 1 7 5 9 1 1 7 7 3 1 1 7 7 5 1 1 7 7 7
1 1 7 9 5 1 1 7 9 7 1 1 7 9 9 1 1 9 1 3 1 1 9 1 5 1 1 9 1
3 1 1 9 3 5 1 1 9 3 7 1 1 9 3 9 1 1 9 5 3 1 1 9 5 5 1 1 9
7 3 1 1 9 7 5 1 1 9 7 7 1 1 9 7 9 1 1 9 9 3 1 1 9 9 5 1 1
1 3 3 1 3 1 3 5 1 3 1 3 7 1 3 1 3 9 1 3 1 5 3 1 3 1 5 5 1

Offline Password Attacks

We will perform one of these in our next lab.

- Frequently much faster (attack speed can scale with attacker resources)
- Can be invisible to defenders (you dont know if/when your password is compromised)
- Many of the same attacks as online (brute force, dictionary, rainbow tables, etc.)
- Requires an offline source to attack
 - File containing stolen password hashes
 - Phishing the user themselves
 - Key logging software that captures the password

‘/etc/shadow’ Exercise

- Create a new user `sudo adduser tempuser`
- Give it a weak password (that you can share with other students)
 - No more than one use of the password: password per table please!!
- Print out that user's salted + hashed password
 - `sudo cat /etc/shadow | grep tempuser`
- Submit that entire output to Pilot (include the user name and all trailing :::
 - Reminder: plain text files only!!!
- I recommend deleting the created user as well:
 - `sudo userdel tempuser`

Authentication defenses

- Password managers (helps ensure length, complexity, AND uniqueness)
- Multi-Factor Authentication (MFA / 2FA)
- Keys/tokens (PKI)
- Biometrics (when used as MFA)
- Policies and procedures

Password Managers

- Allow for much stronger passwords
- Traits of a **stronger** password
 - unique (to you)
 - longer
 - complex
- Convenient for users
 - Until they are **very** inconvenient. . .
- Helps prevent easily guessable passwords
- Helps prevent re-used passwords

Multi-factor Authentication (MFA)

- If passwords are so weak, then we will use another form of authentication alongside them.
- Hopefully a second form of authentication is chosen that is both secure and easy to remember.
- Processes introduced to deal with lost or forgotten MFA can provide attackers avenues of entry or data gathering.

Key based authentication

- Public/Private Key pairs
 - User provides *public key* securely upon account setup
 - User authenticates with *private key*
- Digital Certificates build upon key based authentication
 - Includes digital signature of a certification authority
 - Server verifies credibility of the certificate authority

Relies on unique biological characteristics of the user such as:

- fingerprints
- facial recognition
- speech recognition
- retinal scan
- etc.

Token based authentication

User authenticates and receives a unique encrypted string to use for authentication against other related servers.

Typically used with APIs with multiple frameworks and clients.

Policies and Procedures

- How the people and processes handle all parts of authentication
- How many password attempts?
- How long do you wait for MFA?
- How do you verify a user during account recovery?
- etc.

You (will) get hacked. Then what?

Mat Honan - A case study

- circa 2012
- Wired.com tech blogger
- twitter @mat
- Apple fanboy (joking, but does use apple products)
 - m*****@me.com
- Enjoys amazon.com delivery of goods to his home address

What happened

- August 2012
- 5pm iphone resets
- phone power on and iphone is at setup screen
 - (backups etc were done nightly so no fear yet)
- plug phone in to laptop to restore/recover
 - notification on macbook of incorrect credentials
 - macbook has new (unknown) 4 digit pin protection

What would you do?

What actually happened (all times are approximate)

- Aug 2012
- 4pm buys a toothbrush amazon.com
- 4:30pm amazon account reset with unknown credit card and Mat Honan's address
- 4:45pm me.com reset with last 4 of Mat Honan's CC and address
- 4:47pm gmail.com reset with me.com
- 4:50pm twitter.com reset with gmail.com
- 4:55pm me.com and gmail.com accounts ***permanently deleted*** with all data
- 5pm hacker brags on their newly acquired @mat handle of their hack

The hack

These are the steps the attacker went through:

- First all, the reason behind it, they (attacker) wanted @mat Twitter handle
 - Yes I chose not to update this slide to the new name of X, they are all still tweets in my mind...
 - background research revealed @mat is Matthew Honan
 - find physical address from various online lookups
 - find email address from various online lookups
- try to sign into twitter with that gmail address
 - this confirmed that the gmail address is @mat
- try to sign into that gmail address
 - ***no 2fa !!! :(!!! :(!!!***
 - account recovery is m*****@me.com

The hackening continued

- me.com allows for account recovery with two simple things
 - last 4 of your credit card
 - billing address
- try to sign in with various Mat Honan emails
 - finds the one that works
 - crafts a purchase for a toothbrush *using the hacker's credit card*
 - amazon used to let you purchase save credit card to account without credentials
 - spend money faster!!!
 - buys toothbrush and delivers it to Mat Honan
 - immediately calls amazon, recovers login with hackers credit card info
 - Attacker can now see last 4 of all CC in Amazon as well as billing addresses

Scorched earth

- This lets people into me.com (AppleID)
- which gave them his gmail
- which gave them his twitter
- which was really his entire digital life. . .
- Hacker resets twitter account info, password, and recovery email
- Hacker initiates google account removal, deleting ***ALL google data and account*** (no twitter recovery email anymore)
- Hacker initiates me.com account removal, deleting ***ALL apple data, purchased songs, movies, stored pictures, and removes access to phone and laptop***
- Hacker tweets about his victory

Incident response plan

- Know what ALL forms of authentication are for critical services
- Setup MFA for critical/all accounts
- Know how to disable/re-enable the MFA
- Be prepared to provide necessary information
- Be aware of chained accounts / vulnerabilities

Amazon's fault?

- Yes
- of course
- in hindsight sounded like a great idea